

Programming for Mobile

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Mobile APPs & OSs

Native Mobile Apps: an app you can download and install on your device. A *native mobile app* is developed specifically for a mobile device.

Android

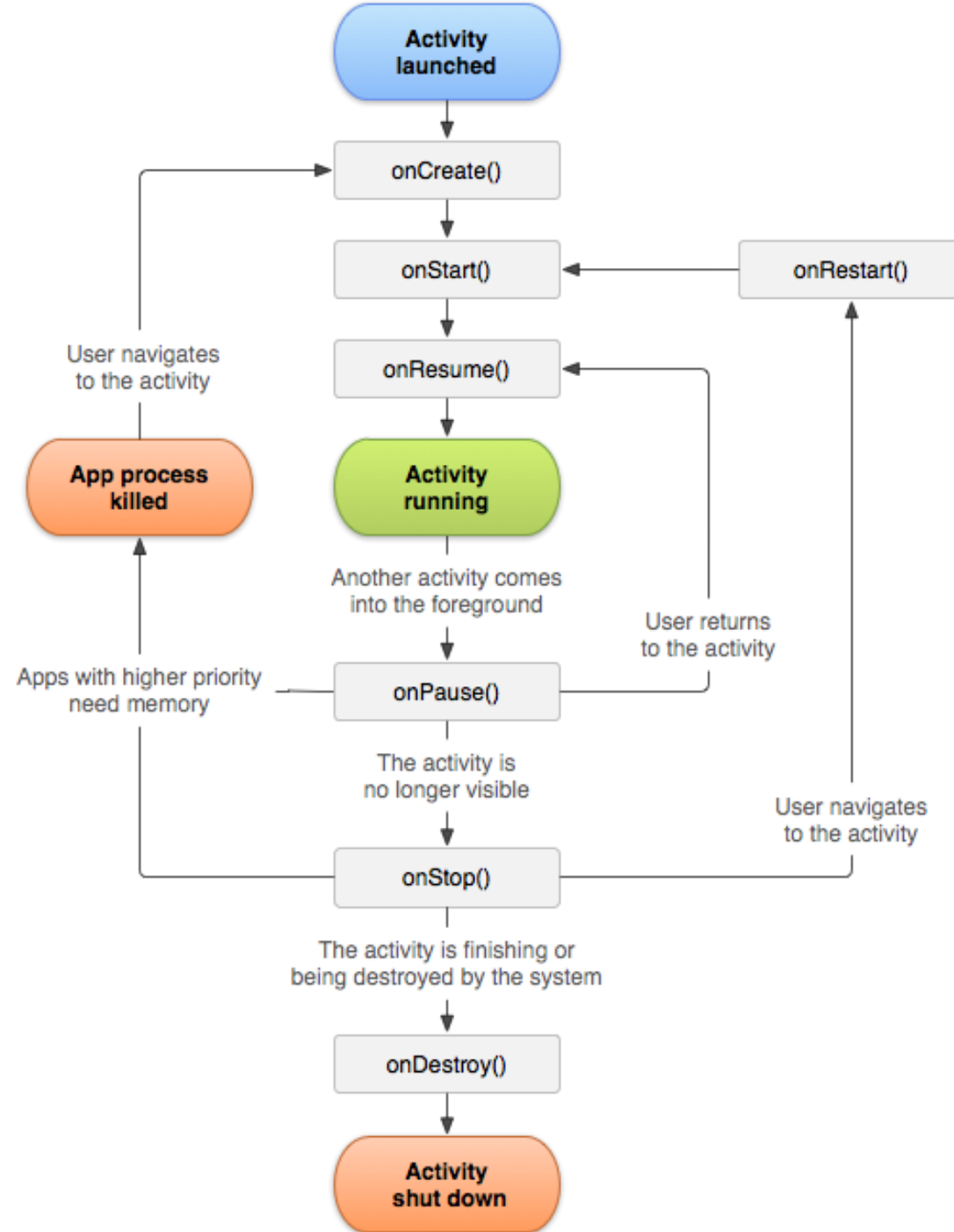
iOS

Web Mobile Apps: not real apps – websites particularly developed and designed for mobile devices – mimic native apps

Web for Mobile → multiplatform but limited in features

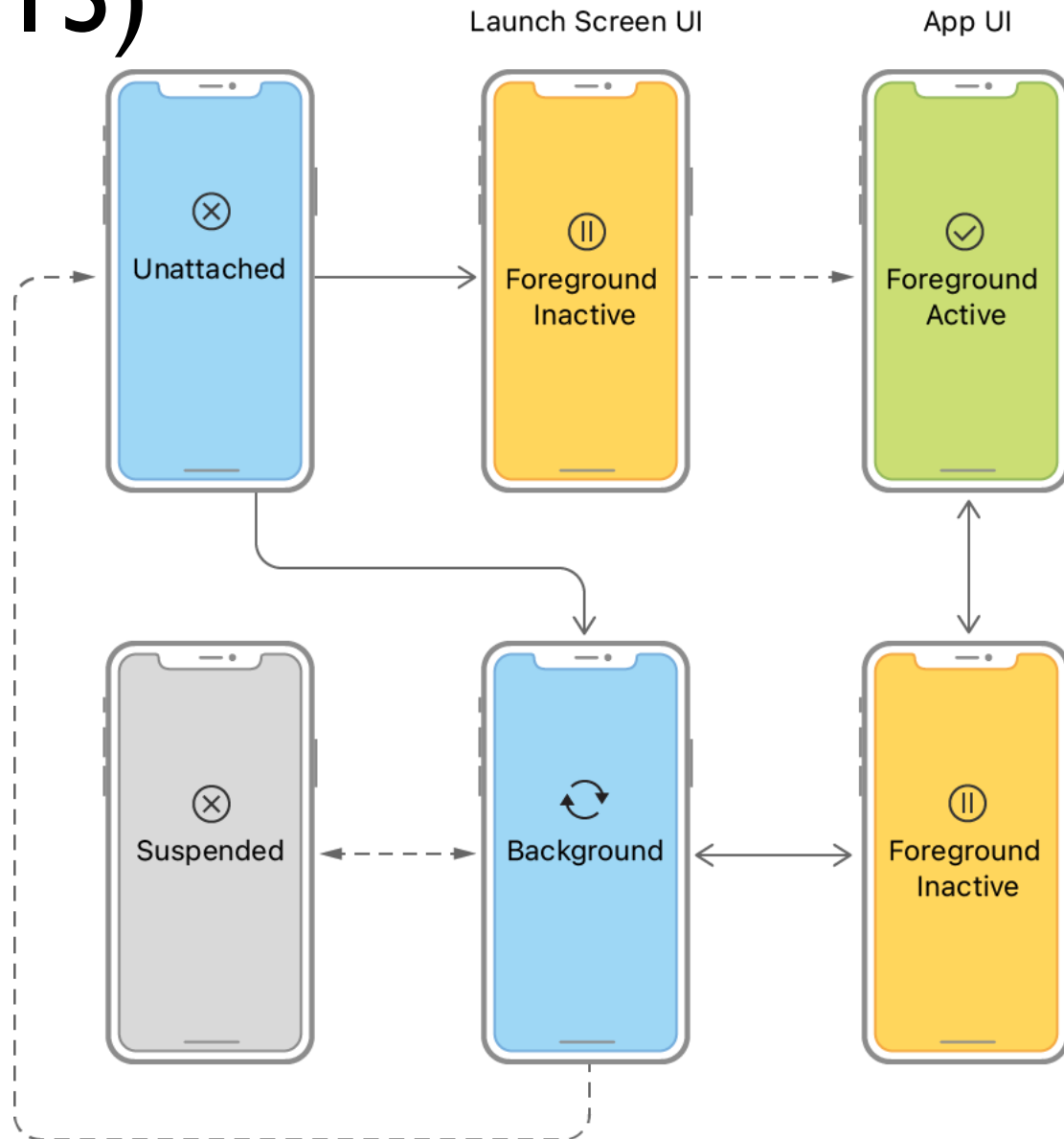
Android

Life-cycle



iOS (≥ 13)

Life-cycle



Mobile APPs

A **hybrid app** is a particular type of native app. Like native apps, users can download and install a hybrid app from app stores. However, the internal structure of native and hybrid apps is very different. Internally, hybrid apps are more like web apps. Hybrid apps lie somewhere between native and web apps.

Like native apps, they live in an app store and can take advantage of the many device features available. Like web apps, they rely on HTML being rendered in a browser, with the caveat that the browser is embedded within the app.

E.g. – Flutter

<https://docs.flutter.dev/resources/architectural-overview>

Crossplatform Frameworks

Write Once, Compile Anywhere

Frameworks to develop native apps for different OSs

- QT: C++ (with QML and Python bindings)

<https://doc.qt.io/qt-6/supported-platforms.html>

- Unity: C#

<https://docs.unity3d.com/Manual/PlatformSpecific.html>